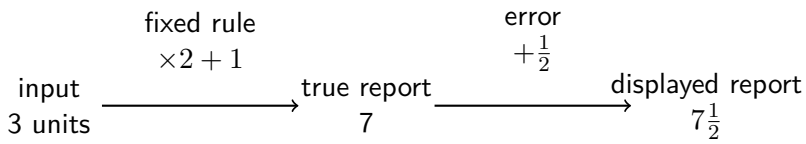
For Problems 6–7, the machine uses one fixed stretch and shift. The reference lengths are exact. Each displayed report may be up to 1 above or below its true report.



Problem 6: For each pair of reference strips, find the smallest and largest possible stretch. Give a machine that reaches each extreme.

Reference pair A	Reference pair B
2 units: display 8 4 units: display 12	2 units: display 8 10 units: display 24

Problem 7: Choose two exact reference lengths from 1 to 10 units before seeing any reports. Which pair gives the smallest guaranteed bound on how wide the range of possible stretches can be, even with the least helpful allowed errors? Could repeating one reference ten times guarantee the same result?